



Chlorophyll Calibration Certificate

RBRtridente bb.chl-a.fDOM, RA180, UW (6000dbar) s/n: 240125
Channel: 2
Calibrated range: 0-50.0 µg/L¹
Initial Accuracy: 5.0% FS (2.5 µg/L)¹

Reference Standards and Materials

Manufacturer	Model/Part No.	Description	Serial No.
Agilent	G9864A Cary 3500	UV-Vis Spectrophotometer, Primary standard	MY2144ZA02
Agilent	G9800A Cary Eclipse	Fluorescence Spectrophotometer	MY2111CG02
Sigma Aldrich	96145	Chlorophyll a	CAS:479-61-8
RBR	RBRtridente / 17400	Three-channel fluorescence and backscatter sensor	233999

Reference method	Scaled Reference Value (µg/L) ¹	Scaled Measured (µg/L) ¹	Scaled Calibration Error (µg/L) ¹	Reference (µg/L) ²	Voltage Ratio, Vr	Measured (µg/L) ²	Coefficients
							C0: -4.667098
							C1: 1.0131155E6
							C2: 103.31017E-3
Black cover	0.000	-0.000	0.000	0.000	0.0000046	-0.000	
Fluorescein solution	49.610	49.610	0.000	480.202	0.0004786	480.202	

Temperature: 21.5 °C

$$Chl_{(\mu g/L)} = (C_0 + C_1 * V_r) * C_2$$

¹ Scaled to the fluorescence response from a monoculture of Thalassiosira weissflogii. (C2 = 103.3107E-3)
² Fluorescence response from the Chlorophyll-a pigment (96145) suspended in acetone. (C2 = 1)

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